Team Foundry

CLAY EARTH DATASET FOUNDRY · TRACK 1 · TEAM 46

An AI agent says it found 15 data centers. Can ADB trust that dataset?

> We built the system that answers it, **with evidence**.

AP Aditya Pasarti

FS Farhan Shahriar

UK Uday Kandi

MP Miguel Pereira

OUR IDEA

Judge the work, not just the answer

Record

Every model and tool call, captured as an OpenTelemetry span.

Verify

Each claim cites its tool result. Code checks the value is there and the site is in the area.

Judge

Three AI judges from three companies score the rest, each vote tied to a step.

The trace is the evidence, not the final answer.

LIVE

Judging a dataset the agent already built

1. **The run:** data centers near Texas parks, built by Kimi K3 on AgentCore
2. **Judge again:** Kimi steps down for its own agent; Mistral sits
3. **While they think:** 15 / 15 claims verified, 8 / 8 sites inside Texas
4. **Verdicts land:** a score, and how far to trust it
5. **Every vote points at evidence:** click a reason, land on the step

▶ Judge this run live

Contrast: a run that fails

Backup: India solar parks

If time: a run an expert reviewed

Scores move a little on each re-judge. Confidence says how much to trust them.

HOW THE SCORE WORKS

Code checks what code can. Judges assess the rest.

Measured by code

Provenance: is each value in the tool result it cites?

Area: is each site inside the boundary asked for?

+

Three AI judges

8 criteria, from tool choice to honest uncertainty.

Opus 5.5

DeepSeek V3.2 · Kimi K3

No judge grades its own family's agent.

→

Median vote

Three votes, no ties: always a real judge's vote.

→

Score + confidence

High · Medium · **Low**: how far the judges agree.

No answer: **FAILED**.

Judges never see the code's results: two independent lines of evidence.

DOES THE EVALUATOR ITSELF WORK?

We sabotaged runs on purpose. It noticed every time.

PLANTED DEFECT	CRITERION IT SHOULD HIT	BEFORE → AFTER
Searches with no area	Tool arguments	2 → 0
A made-up site	Grounding	2 → 0
The answer removed	Completeness	2 → 0

Live, on a 0–2 scale. The first pass caught a bug in our own evidence builder.

Judges agree like experts do

0.50

AGREEMENT SCORE

Human experts in the same kind of study: **0.56**.

Krippendorff's α: 1 = perfect, 0 = chance.
95% CI 0.12–0.67 over 7 runs.

A HUMAN IN THE LOOP

Experts correct the judges, and the judges learn

On one run: reconsider

Expert reviews → Judges re-check the trace → Score revised

Evidence to check, not an order. A judge may keep its vote, and says why.

> "The steps show eight sites were checked against imagery, not three: five Central Valley sites in T6 and three North Coast sites in T9."
>
> Opus, correcting our deliberately wrong review (live, California vineyards)

Across runs: guidance

Reviews → Lessons → Held-out test → A person approves

Approved lessons reach every judge. Versioned on every score, and reversible.

> Tested on a held-out third of reviewed runs it never saw.
>
> Kept per deployment: ADB's judges learn from ADB's experts

Can a review push the judges? Live: ✓ WRONG REVIEW RESISTED ✓ "I INSIST" RESISTED ✓ INJECTED INSTRUCTIONS RESISTED

THE TRACES DON'T LIE

It caught bugs in our own agent

BEFORE

FAILED

Texas parks: no answer, **$1.50** spent. The trace showed why.

AFTER THE FIXES

85.4 · High

Same prompt: 0 errors, 15 / 15 claims, 8 / 8 inside Texas, **$0.81**.

"Northern California" resolved to a 20-metre building, so every search came back empty.

LGND sign-in expired after exactly 300 seconds; every call after 290 s failed.

Narrowed searches around a park were silently widened to all of Texas.

BUILT TO PLUG IN

Every part is swappable by config

Agent

Strands on Amazon Bedrock AgentCore; Claude Sonnet 5.5 or Kimi K3

Tools (MCP)

AWS Open Data (RODA) and LGND Geo imagery search

Trace

OpenTelemetry spans to an S3 archive; exporter swapped by config, proved in CI

Evaluate

Measured checks + judge tribunal on Bedrock; Team Foundry console (FastAPI)

ADB'S RUBRIC → ONE YAML FILE

JUDGES → CONFIG

MODELS → CONFIG

TRACING BACKEND → CONFIG

INFRA → CDK

Any archived run re-scores offline from its trace.

WHAT IT PROVES, AND WHAT'S NEXT

Honest about the limits

It proves today

Claims match their tool output, sites are in the area, and three independent judges agree on the work.

Not yet

Whether a site is **really** a data center. That needs an answer key.

Next

An answer key, ADB's rubric, **reviews from ADB's experts** (the loop is built), plain-language verdicts, full prompt compliance.

Plain-language verdicts and prompt compliance: in progress.

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Thank you. Questions?

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OPEN SOURCE · APACHE-2.0 github.com/onemorepereira/team46-breaking-barriers-hackathon